

# Roof Lifecycle Cost Tool: User Guide

The tool shows a building owner what their roof will cost over the next 20 years in two scenarios: with **PaxSeal planned maintenance**, or with **reactive repair** (fixing problems only after they leak). It turns the difference into dollars, using real cost and storm data for the owner's market.

## Three ways to use it

1

### Custom report

Enter a customer's roof details in the builder and get a branded two-page report to print, save as PDF, or send as a link.

[noah-austin.github.io/TOC-Model](https://noah-austin.github.io/TOC-Model)

2

### Sales library

35 ready-made reports: a typical building of each property type in each market. Open or download, no typing.

[noah-austin.github.io/TOC-Model/library](https://noah-austin.github.io/TOC-Model/library)

3

### Shareable links

Every report has a link with its inputs built in. Send it to a customer or a colleague; it opens straight to the report.

**Markets:** Austin, San Antonio, Hampton Roads, Culbertson (N. Virginia), Laurel. **Property types:** Warehouse / Industrial, Office, Retail, Multi-Family, Mixed Use, Medical / Institutional, K-12 / Education.

## Make a customer report in two minutes

- 1 Customer.** Customer name, property, city and contact. Add your name, phone and email; they print in the report's "Next step" box.
- 2 Roof.** Pick the market and property type, then roof area, age, roof system and condition. Add warranty years left and active leaks if you know them.
- 3 Costs (optional).** Leave blank to use market defaults (shown in gray). Enter the real replacement cost, the PaxSeal price from your proposal, or a cap rate if you have them; the report recalculates instantly.
- 4 Share. Print / PDF** saves the two-page report. **Copy report link** gives a link that opens the finished report without the input form.

**Customer copy vs internal copy.** Reports are customer-ready by default. Under **Model assumptions**, "Show source confidence" makes an **internal review copy** that rates each data source and is marked "Internal review copy" on every page. Don't send that version to customers. The Model assumptions section is for sales ops only.

## What's in a report

**Page 1.** The projected 20-year savings in one sentence, with its range; the property at a glance; net cost for each scenario; the cumulative spending chart; and a roof-life timeline showing when each scenario needs a new roof.

**Page 2.** Where the savings come from (fees, repairs, storm damage, replacements); up to four takeaways for this roof; the local storm risk with FEMA ratings; the key assumptions and their sources; and a next step with your contact details.

How it works

1. Describe the roof

Market, property type, size, age, condition and roof system. Anything left blank uses a market default.

2. Look up market data

Local replacement cost per sq ft, county storm risk from the FEMA National Risk Index, roofing cost growth (4.6% a year, federal price index), and local cap rates by property type.

3. Simulate 20 years, both ways

Each year's fee, repairs, storm damage and any roof replacement is added up for both scenarios. That running total is the chart on page 1.

4. Compare

The difference in net 20-year cost is the projected savings, shown with a conservative-to-upside range.

|              | WITH PAXSEAL                                                             | REACTIVE REPAIR                                                                                   |
|--------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Roof life    | Full design life (20 yrs for TPO, EPDM, mod bit, BUR; 40 for metal)      | About 5 years shorter                                                                             |
| Program fee  | Paid every year, rising 3% a year                                        | None                                                                                              |
| Repairs      | Caught early on scheduled visits; the fee covers half of planned repairs | About 3× the planned rate (more in hospitals, schools and apartments, where leaks do more damage) |
| Storm damage | Expected loss for the county                                             | 1.5× higher with reactive repair only                                                             |
| Replacement  | Later                                                                    | Sooner, and at a higher price because roofing costs keep rising                                   |

Net 20-year cost = cash spent + value of the current roof used up – roof life left at year 20

Counting roof life left at the end keeps the comparison fair: a scenario that buys a new roof in year 19 gets credit for the 19 years it still has. The current roof's value is the same in both scenarios, so it never changes the savings.

Worked example

**San Antonio warehouse:** 100,000 sq ft TPO roof, 8 years old, good condition. A new roof costs about **\$1.10M** today.

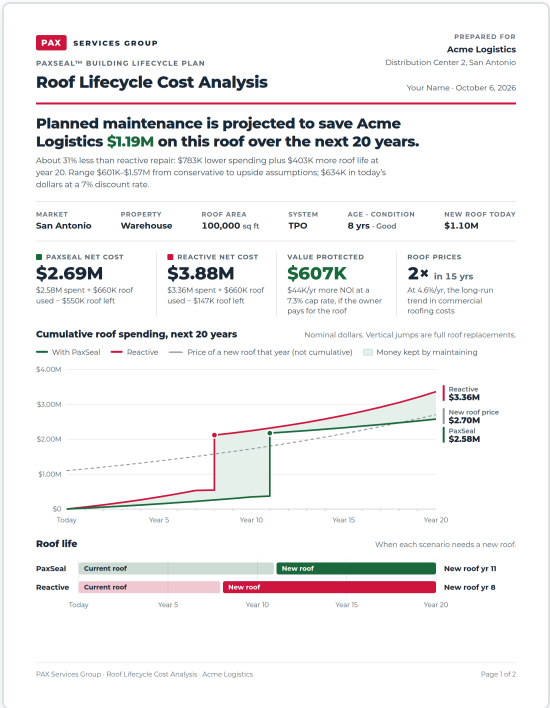
|                   | PAXSEAL | REACTIVE |
|-------------------|---------|----------|
| Next new roof     | Year 11 | Year 8   |
| PaxSeal fees      | \$371K  | —        |
| Repairs           | \$268K  | \$1.58M  |
| Storm damage      | \$138K  | \$207K   |
| Roof replacements | \$1.80M | \$1.58M  |
| Net 20-year cost  | \$2.69M | \$3.88M  |

PROJECTED SAVINGS

\$1.19M

(31%)

\$783K less cash spent, plus \$403K more roof life at year 20. Range \$601K–\$1.57M. Property value protected: \$607K (about \$44K/yr better NOI ÷ 7.3% cap rate).



# Reading the report

| ON THE REPORT          | WHAT IT MEANS                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Projected savings      | How much less the roof costs over 20 years with PaxSeal, split into lower cash spending and extra roof life left at year 20.                                                        |
| Range                  | The same projection under cautious and optimistic assumptions for repair costs and roof life added.                                                                                 |
| Today's dollars        | The savings discounted at 7% a year: what that future money is worth now.                                                                                                           |
| Value protected        | For investor-owned property: yearly operating savings ÷ cap rate. Applies when the owner pays for the roof (not a triple-net lease).                                                |
| Repair & storm savings | Shown instead for schools and hospitals, which are usually owner-occupied: unbudgeted repair and storm costs avoided.                                                               |
| The chart              | Cumulative spending for each scenario. Vertical jumps are roof replacements; the green band is the money kept by maintaining. The dashed line is the price of a new roof that year. |
| What this means        | Up to four takeaways chosen for this roof: fewer roofs or a later replacement, smaller repairs, warranty, and property-type points.                                                 |

## Where the numbers come from

| INPUT                                    | SOURCE                                                                                                         | STRENGTH  |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------|
| Storm risk                               | FEMA National Risk Index, December 2025, by county                                                             | STRONG    |
| Roofing cost growth                      | BLS Producer Price Index for nonresidential roofing, 2007–2025                                                 | STRONG    |
| Roof life with maintenance               | Industry service-life tables; NRC Canada roof study; manufacturer warranty terms                               | MODERATE  |
| Local roof prices, cap rates             | Contractor and broker figures from search summaries; need verification                                         | MODERATE  |
| Repair cost multiplier, base repair rate | Industry estimates; no published study. These drive most of the savings, which is why the report shows a range | WEAK      |
| PaxSeal fee                              | Estimate (\$800 + \$0.13/sq ft a year) until you enter the proposal price                                      | Our price |

## When a customer pushes back

### “Those savings look high.”

Point to the range and to the split between cash and roof value. Even the conservative case is usually clearly positive. The biggest driver is buying fewer or later roofs.

### “Is the PaxSeal fee included?”

Yes. It is a cost every year, rising 3% a year. Enter the actual proposal price for an exact figure.

### “Doesn't insurance cover storms?”

Storm figures are before insurance, and the report says so. Storm damage is a small part of the savings; most comes from repairs and replacements.

### “Our roof is different.”

Enter the real replacement cost, fee or cap rate and the report recalculates on the spot.

## Good to know

**Estimates, not quotes.** Every report says so. The PaxSeal proposal sets the actual price and scope.

**Typical buildings in the library** are illustrative profiles (for example, a 100,000 sq ft warehouse roof, 8 years old), not market averages. Use a custom report for a real property.

**Updates.** The tool is a single web page hosted on GitHub Pages and updates automatically when the model or data change. The full method is in **docs/MODEL\_SPEC.md** in the project repository.